

Week 1 Final Preliminary Proposal

AI-Assisted Early Risk Stratification of Adult Emergency Department Patients Using Routinely Collected Triage Data

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1. Proposed Research Topic

My proposed research topic is **AI-assisted early risk stratification of adult emergency department patients using routinely collected triage data**. This topic focuses on how artificial intelligence can support triage nurses by using information that is already collected when a patient first arrives. This may include age, vital signs, GCS, oxygenation status, FiO₂ where available and the presenting complaint.

The goal is not to replace the triage nurse. Instead, the goal is to explore whether a simple and explainable tool could help nurses identify patients who may be at higher risk of deterioration or urgent clinical need. In a busy emergency department, this type of support could help bring attention to concerning patients earlier while still keeping clinical judgement at the centre of decision-making.

2. PCC Framing

Population: Adult patients presenting to the emergency department.

Concept: AI-assisted early risk stratification using routine triage data.

Context: Emergency department triage, where nurses must rapidly prioritise patients using limited early information.

3. Problem Statement

Emergency departments need a reliable way to identify high-risk patients early but many AI triage tools are difficult to use at the bedside because they rely on complex data or do not clearly explain their decisions. At Mercer General, a 12-week pilot could test an explainable risk stratification tool that uses routinely collected triage data to assign risk levels and show the

clinical reasons behind each flag. The goal is to support nurse decision-making, improve early recognition of high-risk patients and build trust in AI-assisted triage.

4. Search Strategy and Literature Selection

The literature search focused on recent peer-reviewed research related to artificial intelligence, machine learning, emergency department triage, risk stratification and clinical decision support. Google Scholar and PubMed were used because they provide access to healthcare, informatics and clinical AI literature.

Databases searched:

Google Scholar
PubMed

Google Scholar search string:

“artificial intelligence” “emergency department” triage “risk stratification”

PubMed search string:

(“artificial intelligence”[tiab] OR “machine learning”[tiab] OR “AI”[tiab])
AND
 (“emergency department”[tiab] OR “emergency care”[tiab] OR “emergency medicine”[tiab])
AND
 (“triage”[tiab] OR “risk stratification”[tiab] OR “clinical deterioration”[tiab])
AND
 (“clinical decision support”[tiab] OR “prediction model”[tiab] OR “vital signs”[tiab])

Filters and limits applied:

Published within the last five years
English-language articles
Peer-reviewed journal articles
Emergency department triage, risk prediction or clinical decision support focus
Preprints, vendor blogs, opinion pieces, press releases and non-healthcare AI papers were excluded

The final five papers were selected because they each add something important to the proposal. Some explain the broader state of AI in ED triage while others focus more directly on explainability, multimodal triage data and prediction of critical outcomes. Together, they provide a stronger foundation for a practical 12-week pilot at Mercer General.

5. Literature Summaries

Paper 1: Tyler et al. (2024)

Tyler et al. (2024) conducted a scoping review on the use of artificial intelligence and machine learning in hospital emergency department triage. The problem they addressed is that emergency departments are often overcrowded and triage staff must make fast decisions about which patients need urgent care. In this setting, even small delays or triage errors can affect patient safety. The authors searched major databases to explore how AI and machine learning have been used in ED triage and what gaps still exist in the literature.

The review found that AI and machine learning may support ED triage by improving prioritisation, helping with earlier diagnosis and supporting decisions about hospital admission or urgent intervention. This is important because it shows that AI has potential beyond simply sorting patients into categories. It can also support earlier recognition of risk and help clinicians manage crowded ED environments.

However, the paper also highlights an important caution. AI tools may be dangerous if they produce overconfident outputs or if staff trust them without enough validation. This is a key limitation for real-world use because a model that performs well in a study may still fail if it is not properly tested in clinical workflow. The authors also note that more research is needed to understand how AI tools should be safely implemented in ED triage.

This paper is highly relevant to the Mercer pilot because it supports the overall need for AI-assisted triage while also showing why the tool must be explainable, validated and used as decision support rather than automatic decision-making. It helps justify a cautious pilot approach where the system flags risk but the final decision remains with the nurse and clinical team.

Paper 2: Da'Costa et al. (2025)

Da'Costa et al. (2025) reviewed the benefits, challenges and future directions of AI-driven triage in emergency departments. The problem they focused on is that traditional triage can be affected by overcrowding, staff workload, delays and variation in human judgement. Even when nurses use structured triage systems, decisions can still be influenced by pressure in the department, limited time and differences in clinical experience.

The authors explain that AI-driven triage systems may use real-time patient information such as vital signs, medical history and presenting symptoms to support patient prioritisation. Their review suggests that AI could improve patient flow, reduce waiting times, support more consistent decision-making and help allocate resources more efficiently. These points are relevant because Mercer General's proposed pilot is also focused on using early patient information to support faster recognition of risk.

At the same time, the review identifies several barriers that must be addressed before AI triage can be widely used. These include poor data quality, algorithmic bias, lack of explainability, ethical concerns and the need for clinician trust. This is important because an AI tool that gives a risk category without explaining its reasoning may not be helpful to a nurse at the bedside. It may also increase risk if staff do not understand when to question the output.

For the Mercer pilot, this paper supports the need to design a tool that is not only technically functional but also practical and trustworthy. It strengthens the argument that the proposed system should display clinical reasons behind each risk flag such as low MAP, high pulse, low SpO₂ or reduced GCS. This makes the tool more useful for nurse review and more aligned with responsible AI use in healthcare.

Paper 3: Araouchi and Adda (2024)

Araouchi and Adda (2024) proposed TriageIntelli, an AI-assisted multimodal triage system for health centres. The problem they focused on is that triage decisions often require staff to interpret several types of patient information at once. A single vital sign may not tell the full story of a patient's condition. For example, pulse, blood pressure, oxygen saturation and symptoms may all need to be considered together before a patient's urgency becomes clear.

Their system used a multimodal AI approach. This means the model was designed to combine more than one type of patient data rather than depending on a single measurement. This is relevant to triage because nurses naturally assess patients in a multimodal way. They consider vital signs, symptoms, appearance, history and clinical red flags together. A multimodal system therefore has the potential to reflect the way triage decisions are made in practice.

The main value of this paper is that it supports the idea that AI triage tools should not be built around one isolated variable. A tool based only on MAP or only on pulse may miss important clinical context. For example, low MAP with high pulse may be more concerning than either measurement alone. In the same way, low SpO₂ with a high respiratory rate may suggest a different level of urgency than a mildly abnormal value by itself.

The limitation is that a proposed multimodal system still needs proper validation in real clinical settings before it can be trusted. A model may appear strong in design or early testing but still face issues with workflow integration, missing data, staff trust and patient safety. For the Mercer

pilot, this paper supports using routine triage variables together while keeping the prototype simple enough to be tested within 12 weeks.

Paper 4: Lin et al. (2024)

Lin et al. (2024) developed an interpretable deep learning system to identify critical emergency department patients by predicting triage level, hospitalisation and length of stay. The problem they addressed is that ED triage can involve both objective data and subjective judgement. This can contribute to mis-triage, especially when staff are working quickly or when patient presentations are complex. Mis-triage is a serious issue because high-risk patients may wait too long or low-risk patients may receive resources they do not urgently need.

The authors used triage information such as demographics, vital signs and chief complaints. This is especially useful for my proposal because these are the same types of data that may be available early in triage. The model was designed to handle both structured data, such as vital signs and unstructured data such as chief complaint text. This matters because emergency department data is not always clean or purely numerical. Important information may appear in short written notes or presenting complaints.

A major strength of this paper is its focus on interpretability. The authors recognised that ED staff are more likely to accept AI support if the system can explain its output. This directly supports the Mercer proposal because the planned pilot is not just about assigning risk levels. It is also about showing the clinical reasons behind a risk flag. For example, if a patient is labelled high risk, the tool should show whether this was due to low MAP, reduced GCS, abnormal respiratory rate or another concerning pattern.

The limitation is that the model was developed in a specific clinical setting and would need testing before being applied in another hospital. Patient populations, documentation habits and triage workflows may differ between institutions. For Mercer General, this means the pilot should begin as a local feasibility and usability project rather than assuming that a published model can simply be transferred into practice. This paper is one of the most relevant sources for the proposed solution because it connects early triage data, prediction and explainability.

Paper 5: Sitthiprawiat et al. (2025)

Sitthiprawiat et al. (2025) developed and internally validated an AI-based emergency triage model for predicting critical outcomes in emergency department patients. The problem they addressed is that ED overcrowding can delay patient care and traditional triage systems may not always identify high-risk patients consistently. This is directly relevant to emergency care

because triage is not only about order of arrival. It is about recognising which patients are most likely to need urgent attention, ICU admission or increased resources.

The study developed a machine learning model to predict important outcomes such as ICU admission and resource utilisation in ED patients. This is useful because it moves beyond simply predicting a triage category. It focuses on outcomes that matter clinically and operationally. If an AI tool can help identify patients who may need ICU-level care or significant resources, it could support earlier planning and closer review.

The paper is especially applicable to the Mercer pilot because the proposed project is also focused on early risk stratification. The goal is to identify patients who may be at higher risk using data available during triage. This supports the idea that routinely collected ED data can be used to assist with prediction of critical outcomes. It also strengthens the argument that a risk stratification tool should be clinically meaningful rather than only technically impressive.

A key limitation is that the model was internally validated. Internal validation is useful but it does not prove that the tool will work equally well in another hospital or patient population. External validation and real-world testing would still be needed. For Mercer General, this supports a staged approach. The pilot should first test feasibility, data quality, clinical sense-making and nurse feedback before any tool is used to guide real patient decisions.

6. Overall Synthesis of the Literature

Across the five papers, the literature shows that AI has real potential to support emergency department triage but the way it is implemented matters. The strongest theme is that AI can process routine patient data quickly and help identify patients who may need urgent care, admission or closer monitoring. This is valuable in a busy emergency department because nurses often have to make decisions with limited time and incomplete information.

A second major theme is explainability. Several papers show that AI triage tools are more likely to be useful when clinicians can understand the reasoning behind the output. This is especially important in triage because nurses cannot safely rely on a score without knowing what caused it. A tool that says “high risk” should also explain the clinical concern, such as low MAP, high pulse, low SpO₂, abnormal respiratory rate, reduced GCS or a concerning combination of findings.

A third theme is the need for local validation and workflow testing. AI models may perform well in published studies but still fail when moved into a new hospital setting. Differences in patient population, documentation, equipment, staffing and triage workflow can all affect performance. This means the Mercer pilot should not focus only on building a model. It should also test whether the tool is understandable, practical and clinically safe for nurse review.

Overall, the literature supports a cautious and realistic 12-week pilot that uses routine triage data, provides explainable outputs and keeps the nurse in control of the final decision.

7. Gap Analysis

Gap 1: Limited focus on explainable nurse-facing decision support

A common gap in the literature is that many AI triage systems focus heavily on model performance, accuracy and prediction scores. These measures are important but they are not enough for bedside use. In triage, nurses need to know why a patient is being flagged as high risk. A risk score alone may not be helpful if the system does not explain the clinical reasons behind it.

For Mercer General, this gap could be addressed by designing a tool that gives both a risk category and a short explanation. For example, the tool could flag low MAP, high pulse, low SpO₂, reduced GCS, very abnormal respiratory rate or a pattern such as low MAP with high pulse. This would make the output more useful for nurse review and could improve trust in AI-assisted triage.

Gap 2: Need for practical testing using early triage data

Another gap is that many AI triage studies use retrospective datasets or wider electronic health record information that may not be available when the patient first arrives. In real triage, staff often have to make decisions using limited early information such as vital signs, age, GCS, oxygenation status and the presenting complaint.

A Mercer pilot could focus on the first point of assessment by testing whether routine triage data can support early risk stratification. This makes the project more realistic because it does not depend on laboratory results, imaging findings or later physician notes. It also aligns with the actual pressure of ED triage, where decisions need to be made early.

Gap 3: Need for local workflow and trust evaluation

The literature also shows that model performance alone does not guarantee successful implementation. A tool may work well in one hospital but may not fit another hospital's workflow or patient population. If the tool is confusing, too slow or poorly integrated, nurses may not use it even if it is technically accurate.

At Mercer General, the pilot should therefore assess more than prediction. It should also assess whether nurses find the tool clear, useful and practical. This could be done through short

feedback forms, structured usability checks or brief interviews after nurses review sample outputs. This would help determine whether the tool is realistic for local use.

8. Proposed 12-Week Pilot Solution

The proposed solution is a simple explainable AI-assisted triage support prototype for adult emergency department patients at Mercer General. The tool would use routinely collected triage data to assign each patient to a low, moderate or high-risk category. It would also display the clinical reasons behind the risk flag in plain language.

Proposed input variables:

Age
 Gender
 SBP
 DBP
 MAP
 Pulse
 Temperature
 Respiratory rate
 GCS
 SpO₂ where available
 FiO₂ where available
 Presenting complaint where available

Proposed output:

Risk category: Low, moderate or high
 Clinical flags: Short explanation of the concerning findings
 Missing data warning: Notice if important values are missing
 Nurse review prompt: Reminder that the output is decision support only

Example output:

High risk: Low MAP and high pulse detected. Review for possible poor perfusion or clinical deterioration. Nurse assessment required.

This output is intentionally simple. In a triage setting, the nurse does not need a complicated technical explanation. The nurse needs a clear reason why the patient was flagged and what should be reviewed.

9. Pilot Plan and Feasibility

Weeks 1 to 2: Confirm clinical problem and workflow

The team would first confirm the triage problem at Mercer General. This would include reviewing the current triage workflow, identifying what data is routinely collected and speaking with nurses or clinical staff about what type of decision support would actually be useful.

Weeks 3 to 4: Prepare and clean triage data

The team would clean and prepare the triage dataset. This would include checking missing values, validating vital sign ranges, cleaning inconsistent entries and creating derived variables such as MAP where appropriate. This step is important because poor data quality could lead to poor risk classification.

Weeks 5 to 6: Build first prototype

A first version of the risk stratification tool would be created. The prototype could begin as a rule-based or simple machine learning model using routine triage variables. The focus at this stage would be clarity, safety and explainability rather than building the most complex model.

Weeks 7 to 8: Add explanation layer

The prototype would be improved so that each risk output includes a short explanation. For example, if a patient is flagged as high risk, the tool would show whether the concern is low SpO₂, low MAP, reduced GCS, abnormal respiratory rate or a concerning combination of vital signs.

Weeks 9 to 10: Internal testing

The tool would be tested on retrospective triage data. The team would review whether the risk categories make clinical sense, whether the flags are understandable and whether the system

misses obvious high-risk cases. This stage would also check whether missing values affect the output.

Weeks 11 to 12: Nurse feedback and final refinement

The final stage would focus on usability and acceptability. Nurses or clinical reviewers could be shown sample outputs and asked whether the tool is clear, useful and realistic for triage. Feedback would be used to refine the prototype and prepare a final recommendation.

10. Alignment With Mercer General

This pilot aligns well with Mercer General because it is practical, short-term and focused on a real emergency department problem. It uses data that is already collected during triage rather than requiring a completely new workflow. This makes it more feasible for a 12-week project.

The proposed tool also supports the role of nurses. It does not remove clinical judgement or make automatic decisions. Instead, it gives an additional layer of support by highlighting concerning patterns and explaining why a patient may need closer review. This is important because AI tools in healthcare should help clinicians make safer decisions rather than replace them.

The project also builds on the Week 0 data work by connecting cleaned triage variables to a clinical AI use case. Instead of treating data cleaning as a separate technical task, this proposal shows how clean data can support patient safety, early risk recognition and responsible innovation.

11. Expected Benefits

The expected benefits of the pilot include earlier identification of high-risk ED patients, clearer clinical flags for nurse review, better use of routine triage data, improved awareness of missing or abnormal values and a safer foundation for future AI-assisted triage work.

The pilot may also help build trust because the tool would explain its reasoning in plain language. This is important because clinicians are more likely to use a tool when they understand what it is doing and when they can compare the output with their own clinical judgement.

12. Risks and Limitations

The proposed pilot has several limitations. First, the tool would depend on the quality of the triage data. Missing values, incorrect entries or inconsistent documentation could affect the risk category. Second, a 12-week pilot would not be enough to prove that the tool improves patient outcomes. It would mainly test feasibility, clarity, early performance and usability.

Another limitation is that clinical risk is complex. A patient may look stable on paper but still be very unwell based on appearance, pain, medical history or clinical judgement. For this reason, the tool should never replace nurse assessment. It should only support review and highlight possible risk.

There is also a risk of bias if the model is trained on data that does not represent all patient groups fairly. This means the pilot should include attention to data quality, fairness and ongoing human oversight.

13. AI Tool Use Statement

AI tools were used to support organisation of the literature review, refine wording and improve clarity of the problem statement, gap analysis and proposed pilot direction. The final topic, paper selection, interpretation of findings and proposed solution were reviewed and edited by me to ensure that they remained clinically relevant to emergency department triage and aligned with the Mercer General Hospital pilot context. AI was not used as a substitute for reading the selected papers or checking the accuracy of the references.

14. Conclusion

AI-assisted triage has the potential to support emergency departments by helping staff recognise high-risk patients earlier. However, the literature also shows that these tools must be practical, explainable and clinically safe. For Mercer General, the most realistic pilot would be a simple explainable risk stratification tool that uses routine triage data and provides clear reasons for each risk flag.

This proposal is feasible because it focuses on information already collected at triage and keeps the nurse at the centre of decision-making. It also addresses key gaps in the literature by focusing on explainability, early triage data and local workflow testing. Overall, the proposed pilot would be a practical first step toward responsible AI-assisted triage at Mercer General Hospital.

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